

Department of Mathematics, IIT Madras
MA2031-Linear Algebra for Engineers
Assignment-IV

- ✓ 1. Why are the functions $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given below not linear transformations?
(i) $T(a, b) = (1, b)$ (ii) $T(a, b) = (\sin a, 0)$
(iii) $T(a, b) = (|a|, b)$.
- ✓ 2. Does there exist a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $T(1, 1) = (1, -1)$, $T(0, 1) = (-1, 1)$ and $T(2, -1) = (1, 0)$?
- ✓ 3. In each of the following, determine whether a linear transformation T exists. If such a linear transformation exists, then construct one.
(a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$; $T(1, 1) = (1, -1)$, $T(0, 1) = (-1, 1)$, $T(2, -1) = (1, 0)$.
(b) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$; $T(1, 1) = (1, 0, 2)$, $T(2, 3) = (1, -1, 4)$.
(c) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$; $T(1, 0, 3) = (1, 1)$, $T(-2, 0, -6) = (2, 1)$.
(d) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$; $T(1, 1, 0) = (0, 0)$, $T(0, 1, 1) = (1, 1)$, $T(1, 0, 1) = (1, 0)$.
- ✓ 4. Let V and W be real ips. Let $T : V \rightarrow W$ be a linear transformation. Prove that for all $x, y \in V$, $\langle Tx, Ty \rangle = \langle x, y \rangle$ iff $\|Tx\| = \|x\|$.
- ✓ 5. Determine $\text{rank}(T)$ and $\text{null}(T)$ in each of the following cases:
(a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$; $T(a, b) = (a - b, 2b)$.
(b) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$; $T(a, b, c) = (a - b, 2c)$.
(c) $T : \mathbb{P}_2[t] \rightarrow \mathbb{P}_3[t]$; $T(p(t)) = t p(t) + p'(t)$.
- ✓ 6. Let V be the vector space of all functions from $\mathbb{R}$ to $\mathbb{R}$, having derivatives of all orders. Let $T : V \rightarrow V$ be the differential operator: $Tx = x'$. What is $N(T)$?
- ✓ 7. Give an example for each of the following:
(a) A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $N(T) = R(T)$.
(b) Linear transformations $S \neq T$ with $N(S) = N(T)$ and $R(S) = R(T)$.
- ✓ 8. Let U be a subspace of a finite dimensional vector space V .
(a) Does there exist a liner operator T on V such that $R(T) = U$?
(b) Does there exist a liner operator T on V such that $N(T) = U$?
- ✓ 9. Let $T : V \rightarrow W$ be a linear transformation and let $\{v_1, \dots, v_n\}$ be a basis of V . Show the following:
(a) T is one-one iff $T(v_1), \dots, T(v_n)$ are linearly independent.
(b) T is onto iff $\text{span}\{T(v_1), \dots, T(v_n)\} = W$.
(c) T is one-one and onto iff the list of vectors $T(v_1), \dots, T(v_n)$ is a basis for W .
- ✓ 10. Let $T : V \rightarrow V$ be a linear operator such that $T^2 = T$. Let I denote the identity operator. Prove that $R(T) = N(I - T)$ and $N(T) = R(I - T)$.
- ✓ 11. Let $S : U \rightarrow V$ and $T : V \rightarrow W$ be linear transformations where U , V and W are of finite dimensions. Show that $\text{rank}(TS) \leq \min\{\text{rank}(T), \text{rank}(S)\}$.
- ✓ 12. Show that a linear transformation $T : V \rightarrow W$ is an isomorphism iff for every basis $\{v_1, \dots, v_n\}$ of V , the list of vectors $T(v_1), \dots, T(v_n)$ is an ordered basis of W .
- ✓ 13. Let $T : V \rightarrow \mathbb{R}$ be a nonzero linear transformation, where $V \neq \{0\}$. Prove or disprove:
 T is onto iff $\text{null}(T) = \dim(V) - 1$.
- ✓ 14. Give examples of vector spaces V and W , and linear operators $S : V \rightarrow W, T : W \rightarrow V$ such that $ST = I$ but $TS \neq I$.
